

COMPOSITE MATERIAL DATASHEETS
CFRP IM7/8552-UD**MATERIAL IDENTIFICATION**

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Short informal name	IM7/8552-UD
Commercial name	HEXCEL COMPOSITES : HEXPLY 8552/33%/134/IM7 & 8552/33%/268/IM7 (12K)
Specifications	Locals: Z-19.796, Z-19.797 (former Airbus Military / CASA) Airbus: Others: SPE-J-513-E-3025 , SPE-J-513-E-3026 (EFA)
Description	Carbon Fibre tape intermediate modulus (IM7) impregnated in epoxy resin, fiber volume 58%-60%, fibre aerial weights 134g/m ² & 268g/m ² . Service temperature: -55 to 130°C. Cure at 185°C.
Version	0.2 (22/08/2024)

REFERENCES

- [Airbus DS] JST-D-104 Is. 2 "EF2000 Series Production Aircraft CFC Properties and Design Allowables"
- [Airbus DS] NT-J-ID-99037 Is. A "CASA TESTS WORK FOR THE QUALIFICATION OF 8552/IM7 UNIDIRECTIONAL TAPE 0.125 AND 0.250 MM THICK PREPREG MATERIALS TO BE USED ON EF-2000 STRUCTURES (GTP-J-513-E-3026)"
- [NIAR-NCAMP] CAM-RP-2009-015 Rev. B "Hexcel 8552 IM7 Unidirectional Prepreg 190 gsm & 35%RC Qualification Material Property Data Report"

PHYSICAL PROPERTIES

Cured Ply Thickness (CPT)	0.125 0.250 mm	Coefficients of Thermal Expansion CTE (1/°C)	Range	-55...20°C	20...70°C	70...120°C
Density	1590 kg/m ³		α_1	-1.0E-6	-1.7E-6	-1.7E-6
Tg-onset (wet)	>130 °C		α_2	2.6E-5	3.1E-5	3.1E-5
Max. Operational Limit (MOL)	130 °C		α_3	2.6E-5	3.1E-5	3.1E-5

ELASTIC MODULI

Property / Units	E_1 (MPa)	E_2 (MPa)	G_{12} (MPa)	ν_{12} -	K_B -
Values	155000	8400	3200	0.3	0.90

Remarks:

- The K_B is the buckling correction factor for moduli to apply to E_1 , E_2 and G_{12} in buckling stability analyses

Property / Units	E_3 (MPa)	G_{13} (MPa)	G_{23} (MPa)	ν_{13} -	ν_{23} -
Values	-	-	-	-	-

PLAIN STRENGTHS

Ultimate stresses of the ply (from UD-laminates testing).

Property	Mean Values (MPa)				B-basis K_B	E -KDF			B-basis Values (MPa)			
	RT/Dry	70°C/W	100°C/W	130°C/W		70°C/W	100°C/W	130°C/W	RT/Dry	70°C/W	100°C/W	130°C/W
F_{1t}	2600	-	2600	2500	0.84	-	1.000	0.962	2184	-	2184	2100
F_{1c}	1500	-	1250	1050	0.86	-	0.833	0.700	1290	-	1075	903
F_{2t}	60	-	30	25	0.80	-	0.500	0.417	48	-	24.0	20.0
F_{2c}	290	-	175	150	0.73	-	0.603	0.517	211.7	-	127.7	109.4
F_{12}	90	-	80	70	0.94	-	0.889	0.778	84.6	-	75.2	65.8

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Design values for checking the laminate strength using the Maximum Truncated Strain Criterion.

Property	Mean Values ($\mu\epsilon$)				B-basis K_B	E-KDF			B-basis Values ($\mu\epsilon$)			
	RT/Dry	70°C/W	100°C/W	130°C/W		70°C/W	100°C/W	130°C/W	RT/Dry	70°C/W	100°C/W	130°C/W
ϵ_{tall}	-	-	-	-	-	-	-	-	15400	-	14100	13500
ϵ_{call}	-	-	-	-	-	-	-	-	8300	-	7000	5800

INTERLAMINAR PLY STRENGTHS

Property	Mean Values				B-basis K_B	E-KDF			B-basis Values			
	RT/Dry	70°C/W	100°C/W	130°C/W		70°C/W	100°C/W	130°C/W	RT/Dry	70°C/W	100°C/W	130°C/W
F_{13} (MPa)	117	-	64	58	0.87	-	0.547	0.496	101.8	-	55.7	50.5
F_{3t} (MPa)	-	-	-	-	-	-	-	-	-	-	-	-
G_{IC} (N/mm)	-	-	-	-	-	-	-	-	-	-	-	-
G_{IIC} (N/mm)	-	-	-	-	-	-	-	-	-	-	-	-

OPEN HOLE AND BOLTED JOINTS

Reference material strengths in open-hole, filled-hole and bearing for the quasi-isotropic laminate.

Property	Mean Values (MPa)				B-basis K_B	E-KDF			B-basis Values (MPa)			
	RT/Dry	70°C/W	100°C/W	130°C/W		70°C/W	100°C/W	130°C/W	RT/Dry	70°C/W	100°C/W	130°C/W
OHT_{QI}	400	-	-	-	0.87	1.0	1.0	1.0	348.0	348.0	348.0	348.0
OHC_{QI}	340	-	-	-	0.83	-	0.83	0.70	282.2	-	234.2	197.5
FHT_{QI}	450	-	-	-	0.87	1.0	1.0	1.0	391.5	391.5	391.5	391.5
FHC_{QI}	480	-	-	-	0.83	-	0.83	0.70	398.4	-	330.7	278.9
F_{BR-QI}	800	-	-	-	0.83	-	0.83	0.70	664.0	-	551.1	464.8
F_{PO}	-	-	-	-	-	-	-	-	-	-	-	-

Notes:

- B-basis correction factors from typ. populations CV values of fibrous composites : 5.8% (fibre driven properties) and 9% (resin driven properties)
- EKDF factors 1.0 for fiber driven properties (tension), copied from related lamina strengths
- FHT for the QI laminate close in JST-D-104 Fig. 4-1 (if it were mean value) and NCAMP datum
- FHC for the QI laminate close in JST-D-104 Fig. 4-2 (if it were mean value at 130°C/W condition) and NCAMP datum
- OHT taken from NCAMP report, more conservative JST-D-104 Fig. 4-3
- OHC for the QI laminate close in JST-D-104 Fig. 4-4 (if it were mean value at 130°C/W condition) and NCAMP datum
- FBR taken from the NCAMP report, where test was done in single-shear configuration

COMPOSITE DAMAGE TOLERANCE

Design values for residual strengths after BVID impact in tension and compression.

Property	B-basis Value for all conditions
TAI ($\mu\epsilon$)	5500

Property	Thickness	B-basis Value for all conditions
CAI ($\mu\epsilon$)	$0.75 \text{ mm} \leq t \leq 4 \text{ mm}$	2600
	$4 \text{ mm} \leq t \leq 8 \text{ mm}$	3600
	$t \geq 8 \text{ mm}$	5500

(*) Impact energy up to 30J